

BijliSaathi — Hackathon Product Spec

2026-09-18 · @Someone

Summary

BijliSaathi (working name) is a WhatsApp assistant that sends an Indian family one short plan a day. It says which appliances to run when, to cut the bill, ride out tonight's power cuts, and take load off the evening peak.

Every message delivers one thing: **shift this appliance, to this time, save this much.**

Why it can win:

- **It fits the theme head-on.** It starts from a problem the team's own families face: summer bills and night-time power cuts.
- **Judges take part.** A judge scans a QR code, sends a bill photo and gets their own plan live.
- **Real data, honestly labelled.** Grid, weather and tariff data are real; every estimate is marked as one.
- **A clean AWS story.** Bedrock, Lambda, EventBridge, Step Functions, DynamoDB and Amplify each do one clear job.
- **A national story told through one home.** Millions of small shifts are how India gets through its evening peak.

Primary track: **Deployed, with a URL.** Open-source pieces (Strands Agents, Cedar) are stretch goals.

The problem

India has enough electricity on paper, yet families lose power on hot evenings and pay the most for it then. The squeeze lands exactly when homes need power most.

Personal. On summer nights the AC, fans and fridge all run hardest. That is when bills climb and cuts hit. Open question: what is the team's own story — which outage or bill made this personal?

Community. Every home in a colony or apartment block switches on its AC in the same few evening hours. Shared transformers and feeders overload at once, and whole streets go dark together.

Country. The national numbers show the same gap between total supply and evening reality.

Fact (2026)	Figure	Source
Peak demand met	270.8 GW, May 2026	Energetica India
Energy shortage, FY 2025-26	Zero, nationally	Energetica India
Coal and lignite share of supply, Apr–Jun 2026	69.54%	Energetica India
Coal in non-solar peak hours	Up to 188.8 GW, about 75% of 251.4 GW	Energetica India
Evening peak deficit, May heatwave	About 2.57 GW (Grid-India)	Reuters, via The News
Night outages reported	40–60 min cuts in parts of Chennai; complaints in Delhi, Noida, Odisha	Reuters, via The News

Supply is sufficient in total but fails locally and in the evening, when coal carries most of the load.

Core insight: timing, not total

India's evening problem is a timing problem. Solar floods the grid at midday; coal carries the evening. The cheapest, cleanest power plant India can build is 8 pm usage moved to 1 pm.

- **Midday:** solar supplies a large share, so power is cheaper and cleaner.
- **Evening:** coal supplied up to about 75% of generation in non-solar peak hours in Apr–Jun 2026, and deficits appear.
- **Growth:** every new AC and EV adds to the evening unless its use moves earlier.

The government already prices this. Under the Time-of-Day (ToD) rules, the eight solar hours set by each state regulator must be 10–20% cheaper than normal, and peak hours 10–20% dearer ([SolarQuarter](#)). This applies to all non-agricultural consumers from 1 April 2025 at the latest, and immediately once a smart meter is installed. The same rules require smart meters to be read remotely at least daily, with the data shared with consumers.

The gap: rollout varies by state, and most families don't know their cheap hours exist. Nobody tells them, in their language, what to move and what it saves. That is the job BijliSaathi does.

Open question: which state will the demo use, and does its current tariff order bill residential ToD rates?

Who it's for

The primary user is the person who runs a city or town household's electricity decisions and already lives on WhatsApp. Everything in the demo serves this one person.

User	What they need	In the hackathon build
Household decision-maker (primary): home with AC, geyser, water pump, washing machine, and/or an EV or e-scooter	When to run what, a lower bill, a warning before cuts	Yes — the whole product
Other family members	A simple instruction they can follow without an app	Yes — shared plan, regional language
Apartment society / RWA manager	Keep shared load and EV charging under limits	Stretch only
DISCOM or policymaker	Aggregate peak relief from many homes	Vision slide only

Not for, in this build: agricultural consumers (excluded from ToD rules), industrial and commercial users, and homes without WhatsApp (a voice channel is a later idea).

The experience

A family goes from "Hi" to their first personal plan in under two minutes, with no app to install. After that, BijliSaathi sends at most two messages a day.

flowchart TD

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A[Scan QR or say Hi] --> B[Pick language]
B --> C[Share pincode]
C --> D[Send bill photo]
D --> E[Tap appliances owned]
E --> F[Spending breakdown]
F --> G[First plan for tomorrow]
G --> H[Daily loop]
H --> I[Afternoon: cut-risk alert]
H --> J[Evening: tomorrow's plan]
H --> K[Ask: Can I run X now?]
J --> L[Tap Done to log a shift]
L --> M[Weekly savings report]

```

Onboarding collects only what the plan needs: language, pincode (for tariff and weather), one bill photo and the appliance list.

Message	When	What it says
Tomorrow's plan	Every evening	Cheap, clean window; top three shifts; rupee estimate for each
Cut-risk alert	Afternoon, high-risk days only	Tonight's risk window and what to do before it
Answer	On demand	"Can I run the washing machine now?" with a one-line reason
Weekly report	Once a week	Rupees saved, kWh moved out of the peak, CO ₂ avoided, neighbourhood total

An illustrative evening message (numbers are placeholders):

Tomorrow (Sat) — cheap hours 9 am–5 pm

1. Scooter: charge 10 am–1 pm, not 9 pm → saves ₹X
2. Washing machine: run before 2 pm → saves ₹X
3. Water pump: run at 11 am

Tonight: cut risk HIGH, 8–11 pm. Pre-cool the bedroom by 7 pm and charge phones. Reply DONE after each, or ask me anything.

A small web page on the deployed URL shows the family's breakdown and savings, plus a public impact counter. WhatsApp stays the main surface.

Core features (must build)

Nine features make the product; each one feeds the single message "shift this, to this time, save this much." Anything that doesn't feed it is cut.

#	Feature	What it does	Why it matters
1	WhatsApp onboarding	Language, pincode, appliance list in under two minutes	No install; works for every family member
2	Bill reader	Reads a bill photo: units, amount, billing period, tariff category, meter type; user confirms	Personalises from real numbers with zero typing
3	Appliance spending breakdown	Estimates each appliance's share, scaled to match the bill total; shown as ranges	Tells the family where the money goes ("AC is about 40%")
4	Cheap and clean hours	Tomorrow's best window from the state's ToD hours plus forecast solar supply	The core signal every plan is built on
5	Cut-risk alert	Tonight's risk (low / medium / high) from heat forecast, grid stress and local reports	The reason families read the messages in summer
6	Daily shift plan	Top three actions, ranked by rupees saved and evening-peak relief	The one message the product exists to send
7	Ask anytime	Answers "Can I run X now?" with a one-line reason	Makes it a conversation, not a broadcast
8	Savings and impact tracker	Logs DONE taps; totals rupees, kWh moved out of the peak, CO ₂ avoided	Proof for the family and for the judges

#	Feature	What it does	Why it matters
9	Public impact counter	Web page with all users' kWh moved out of the peak, plus a labelled scale projection	Connects one home to the national evening peak

Shiftability rules. The plan only suggests what can really move, and says so when something can't.

Appliance	Shiftable?	Suggested move
EV / e-scooter	Yes — the biggest win	Solar hours if parked at home; otherwise after the evening peak
Water pump	Yes	Fill the tank at midday
Washing machine, iron	Yes	Inside the solar window
Geyser	Partly	Heat just before use; avoid the evening peak
AC	Partly	Pre-cool before the peak; raise the set-point during it
Fridge, fans, lights, Wi-Fi	No	Never suggested

EV module — the hero feature

The EV is the largest and most flexible load in a home, and it is the demand growth that started this idea. It gets its own module and leads the demo.

Asked once at setup: vehicle type (e-scooter or car), charger type, typical daily kilometres, where it's parked in the daytime (home, office, varies), usual departure time, and whether the office has a charger.

flowchart TD

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A[Tomorrow's charging need] --> B{Parked at home<br/>in solar hours?}
B -->|Yes| C[Charge in solar window<br/>cheapest + cleanest]
B -->|No| D{Charger at office?}
D -->|Yes| E[Charge at office<br/>in daytime]
D -->|No| F[Charge after evening peak<br/>cheaper, not cleaner]
C --> G[Top up only what<br/>tomorrow's km need]
E --> G
F --> G

```

The nuance most teams will miss: a commuter's car is rarely home during solar hours. Charging after the evening peak saves money and eases the crunch, but night power is mostly coal. The product says this plainly and never labels night charging as green.

The end of the evening peak differs by state and season, so it is set per state rather than as a fixed clock time.

Approximate loads for the plan: home car chargers draw a few kilowatts (roughly 3–7 kW); e-scooter chargers usually draw under 1 kW.

The hackathon version advises; it does not control the charger. Most Indian EVs and home chargers don't expose open controls, so automatic charging is a vision item.

Data: what's real, what's estimated

Every number on screen and on every slide carries a label: Real, Estimated, User-reported or Projection. Judges forgive estimates; they punish fake precision.

Data	Where it comes from	Label	Note
State demand and generation mix	Grid-India published reports	Real	Confirm the time resolution you can download before building on it
Weather forecast (temperature, heat, cloud)	IMD, Open-Meteo	Real	Drives both solar supply and evening load
ToD hours and rates	Demo state's tariff order	Real	Entered by hand for one or two states

Data	Where it comes from	Label	Note
Household bill	User's bill photo	Real, user-provided	User confirms the reading
Appliance-level split	Appliance list, calibrated to the bill	Estimated	Shown as ranges, never exact
Hourly carbon signal	Derived from the generation mix	Estimated from real data	Used for "cleaner hours," not precise CO ₂ claims
Tonight's cut risk	Heat, grid stress, local reports	Modelled	A risk window, not a prediction of a specific cut
Outage reports	Users tapping "power's out"	User-reported	Stretch feature
Shifts done	DONE taps	User-reported	Savings are "as logged"
Scale projection	Arithmetic on logged shifts	Projection	Always labelled on the counter
May 2026 heatwave backtest	State-level historical data	Real	The proof slide

Where a family can export their smart meter's daily data, it replaces the estimate. That is a stretch path, not a dependency.

Intelligence layer

One rule runs through all of it: **the model writes the words; the calculator writes the numbers**. The language model never invents a rupee figure, a time window or a risk level.

Component	Method	Output
Bill reading	Bedrock multimodal model extracts fields into a fixed schema; user confirms	Units, amount, period, tariff category, meter type

Component	Method	Output
Appliance estimate	Rated power × hours × usage factor (star rating, AC duty cycle, season), scaled to match the bill's units	Share of bill per appliance, as a range
Cheap and clean hours	State ToD window intersected with tomorrow's forecast solar supply (cloud cover)	Best 2–4 hour window for tomorrow
Demand and stress forecast	State demand with temperature as an input; Amazon's open-source Chronos time-series models vs a simple temperature-regression baseline	Tomorrow's evening stress level
Cut-risk score	Rules on forecast demand vs recent peak met, heat index, recent deficit days, local reports	Low / medium / high, with a time window
Plan ranking	Shiftable kWh × (peak price – cheap price), plus a peak-relief weight; respects at-home hours and departure times	Top three actions only
Message writer	Bedrock turns the ranked plan into the user's language, eight lines at most	The evening WhatsApp message
Question answering	Bedrock calling tariff, window and plan lookups	One-line answer with the reason

For the forecast, keep whichever model wins the backtest and report its error honestly. If the simple baseline wins, say so; that reads as rigour, not weakness.

AWS architecture

The build targets the **Deployed, with a URL** track: fully serverless, so it scales to zero when idle. Two open-source pieces are optional extras for the other track.

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flowchart LR
    U[Family on WhatsApp] --> W[End User Messaging<br/>WhatsApp channel]
    W --> L[Lambda: chat handler]
    L --> B[Bedrock: bill + words]
    L --> D[(DynamoDB)]
    S[EventBridge Scheduler] --> SF[Step Functions<br/>daily pipeline]
    SF --> F[Fetch grid + weather]
    F --> S3[(S3 raw data)]
    SF --> M[Forecast, risk, plans]
    M --> D
    SF --> W
    WEB[Amplify web page] --> API[API Gateway]
    API --> D
```

Service	Job in BijliSaathi
AWS End User Messaging (WhatsApp)	Sends and receives the family's messages
API Gateway + Lambda	Chat webhook and the web page's API
EventBridge Scheduler	Triggers daily data pulls and message sends
Step Functions	Daily pipeline: fetch → forecast → risk → plans → send
DynamoDB	Households, appliances, plans, shift logs, outage reports
S3	Raw grid and weather data
Bedrock	Bill reading, message writing, question answering
Chronos (open source)	Demand forecast, run as a pipeline step
Amplify Hosting	The deployed URL: family page and impact counter
Cognito	Optional family login for the web page
Strands Agents SDK (stretch)	Question-answering agent for the open-source track
Cedar (stretch)	Who sees what: family, society manager, public totals

Privacy by design: keep only the extracted bill fields, never the consumer number, and delete the bill image after reading it.

Open question: confirm WhatsApp Business setup early, since verification can take time. The fallback is a Telegram bot or web chat on the same backend.

Stretch features, in priority order

Build these only after all nine core features work end to end. Take them strictly top-down.

1. **"Power's out" reports and a live pincode map.** One tap reports a cut; the map updates and feeds the cut-risk score. It also makes a strong live demo moment.
2. **One live-metered appliance on stage.** An off-the-shelf smart plug with energy monitoring and a local API, powering a kettle or fan. Switch it on and the dashboard moves. No DIY wiring into mains.
3. **Society / RWA mode.** One plan for an apartment block, staggering EV charging to stay under the shared load limit.
4. **Smart meter data import.** Where a family can export daily readings, they replace the appliance estimate.
5. **Voice notes in regional languages.** For family members who won't read a long text.
6. **Rooftop-solar homes.** "Use your own solar first" plans for homes with panels.
7. **Strands agent and Cedar roles.** Credit on the open-source track, if the rules allow using both.

Out of scope

These ideas came up and are deliberately cut. Each one would dilute the single message or depend on data the team can't get.

Idea	Why it's out
Controlling appliances (smart-home platform)	Needs hardware in every home; Alexa, Google Home and smart-plug brands already own this
Metering every appliance individually	Needs a device per appliance; consumer meter data is too coarse to split reliably

Idea	Why it's out
National long-range demand forecasting	The Central Electricity Authority and Grid-India already do it; there is no user to demo to
District transformer-level prediction	Needs DISCOM feeder and transformer data the team can't access; kept as a vision item
Agricultural pumps	Excluded from ToD rules; a different problem with different fixes
Bill payments	Adds compliance work and nothing to the core message
A native mobile app	WhatsApp removes install friction entirely
General energy dashboard with graphs of everything	Fails the "shift this, save this" test

Demo script

The demo's centre is a judge using BijliSaathi on their own phone. The backtest is the proof; everything else supports those two moments.

1. **Hook.** The team's own outage or bill story, then the paradox: India met 270.8 GW in May, yet homes lost power at night.
2. **Live onboarding.** A judge scans the QR code, picks a language, photographs a printed sample bill and taps their appliances. Their breakdown and tomorrow's plan arrive in under two minutes.
3. **EV moment.** The judge asks, "Can I charge my scooter now?" The answer names the window and the reason.
4. **Outage moment (if built).** Two or three team phones tap "power's out." The map lights up, risk turns high, and an alert goes out.
5. **Proof.** One slide replays the May 2026 heatwave: what BijliSaathi would have sent a day or two earlier, and the evening deficit window it flagged.
6. **Impact.** The counter shows kWh moved by everyone who used it during judging, beside a clearly labelled projection.

7. **Close.** One line of vision: millions of homes moving a little load is how India gets through its evening peak.

Bring to the stage:

- Printed sample bills from the demo state, with personal details removed
- A backup phone already onboarded
- A recorded video of the full flow
- Your own mobile hotspot

Pitch narrative and slide flow

The pitch moves from one home to the whole country, then back to the phone in the judge's hand. Eight slides, with the live demo as slide five.

#	Slide	The one message	On screen
1	The moment	Our street goes dark at 9 pm in summer	The team's own photo or one sentence
2	The paradox	Enough power on paper, cuts at night	270.8 GW met; zero shortage nationally; 2.57 GW evening deficit
3	The insight	It's timing, not total — and ToD already pays for shifting	Midday solar vs evening coal (about 75% in non-solar peak hours)
4	The product	One message a day: shift this, save this	The WhatsApp message itself
5	Live demo	A judge uses it now	Phone mirrored to the screen
6	How it works	The calculator writes numbers; the model writes words	Architecture diagram with Real / Estimated labels
7	Proof	It would have warned families before the May heatwave	Backtest chart, plus any test-family results
8	Scale	Homes → societies → DISCOM demand response	Impact counter and the vision line

Tagline options:

- "Shift the hour, not the lifestyle."
- "The cheapest power plant is 8 pm moved to 1 pm."

Hard judge questions

Every question below has a short, honest answer ready. Conceding a limit plainly scores better than defending it.

Question	Answer
Isn't this just an energy-saving tips app?	Tips apps are static. This uses live grid, weather and tariff data, the family's own bill and appliances, and warns about tonight's cuts.
How much does a family actually save?	It depends on the state's ToD rates and the appliances. Solar hours must be 10–20% cheaper and peak 10–20% dearer; AC and EV homes gain most. We compute it from their own bill.
What if their state doesn't bill residential ToD yet?	The value shifts to cut warnings and cleaner hours. The plan still eases the evening peak.
Your appliance numbers are guesses.	Yes: estimates calibrated to the real bill total, shown as ranges. Smart meter data or a smart plug replaces them.
Can you really predict outages?	We predict risk windows, not specific cuts. The state-level stress and heat signals are real; local precision grows with user reports.
Charging at night isn't green.	Correct, and the product says so. It still eases the evening crunch; clean charging needs daytime parking at home or office.
Why WhatsApp?	It's already on almost every phone in India, needs no install, and works in regional languages.
Does anyone actually use it?	Open: test with a few real families and quote them in the pitch.

Question	Answer
What about privacy?	Only extracted bill fields are stored, never the consumer number; the image is deleted after reading.
Why won't DISCOMs just do this?	They are busy installing meters and fixing billing. BijliSaathi could become their demand-response channel to households.
How is this different from Pravāh?	Pravāh builds forecasting for utility control rooms. BijliSaathi works on the household side, so the two are complementary.

Risks and mitigations

The biggest risks are a channel that won't connect, data that isn't granular enough, and scope creep. Each has a fallback decided in advance.

Risk	Impact	Mitigation
WhatsApp Business setup is slow	High	Set it up first; keep a Telegram bot or web chat on the same backend as a fallback
Grid data lacks hourly detail	High	Check resolution first; otherwise use daily reports plus a typical daily load shape, labelled Estimated
Scope creep	High	Apply the "shift this, save this" test to every feature
Overclaiming numbers	High	Real / Estimated / Projection labels on every screen and slide
Demo network or live failure	High	Own hotspot, pre-onboarded backup phone, recorded video
State tariff rules are complex	Medium	Support one or two states, entered by hand
Bill photo misread	Medium	A confirmation step; test on sample bills from the demo state

Risk	Impact	Mitigation
Forecast is weak	Medium	Show the baseline honestly; frame the output as a risk level

Definition of done and success metrics

The build is done when a stranger can go from "Hi" to a correct personal plan on stage, and every number shown is labelled.

- ☐ A new user gets a personal plan in under two minutes
- ☐ Sample bills from the demo state are read correctly
- ☐ Tomorrow's plan is generated from live grid, weather and tariff data
- ☐ Tonight's cut-risk message is generated from live data
- ☐ "Can I run X now?" gets a correct answer with a reason
- ☐ DONE taps update the savings tracker
- ☐ The web page and impact counter are live on a public URL
- ☐ The May 2026 heatwave backtest is complete, with a chart
- ☐ Every number on screen carries a Real / Estimated / Projection label
- ☐ A backup video of the full flow is recorded

Numbers to measure and quote in the pitch:

Metric	How it's measured
Onboarding time	Stopwatch on first-time users
Bill-reading accuracy	Correct fields across the sample bills
Forecast error	Model vs baseline on the backtest
Test families	Count, plus one quote from a real family
kWh moved out of the peak	Sum of logged shifts during testing

Vision beyond the hackathon

The long game is millions of homes acting together like a power plant that switches on at the evening peak. That answers the original worry: India's demand will keep growing, and shifting when we use power is the cheapest way to meet it cleanly.

Stage	What it adds	Who might pay (ideas)
1. Households	Daily WhatsApp advice (this build)	Free for families
2. Apartment societies	Shared-load plans and staggered EV charging	Societies and RWAs
3. Automatic shifting	Smart plugs, smart chargers, smart meter data	Device and charger partners
4. DISCOM demand response	Verified peak reduction from enrolled homes	Utilities, instead of buying costly evening power

Stage 4 is where BijliSaathi meets utility-side tools like Pravāh's: they forecast the peak, and households help flatten it.